

EEE 554 Homework 4

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Problem 1

Problem Statement: The random variable x has the Erlang density $f(x) \sim c^4 x^3 e^{-cx} U(x)$. We observe the samples $x_i = 3.1, 3.4, 3.3$. Find the ML estimate $\hat{c}$ of c .

Solution. The probability density function for the Erlang random variable is $f(x, c) = \frac{c^4}{3!} x^3 e^{-cx}$. For n independent samples, the likelihood function is:

$$f(X, c) = \prod_{i=1}^n \frac{c^4}{6} x_i^3 e^{-cx_i} = \left(\frac{c^4}{6}\right)^n \left(\prod_{i=1}^n x_i^3\right) e^{-c \sum_{i=1}^n x_i}$$

The log-likelihood function $L(c)$ is:

$$L(c) = n(4 \ln c - \ln 6) + 3 \sum_{i=1}^n \ln x_i - c \sum_{i=1}^n x_i$$

Setting the derivative with respect to c to zero for the maximum:

$$\frac{dL(c)}{dc} = \frac{4n}{c} - \sum_{i=1}^n x_i = 0 \implies \hat{c} = \frac{4n}{\sum_{i=1}^n x_i}$$

Given the samples 3.1, 3.4, 3.3, we have $n = 3$ and a sum of 9.8. Substituting these values:

$$\hat{c} = \frac{4(3)}{9.8} = \frac{12}{9.8} \approx 1.2245$$

The ML estimate $\hat{c}$ is approximately 1.2245. □

Problem 2

Problem Statement: The random variable x has the truncated exponential density $f(x) = ce^{-c(x-x_0)} U(x-x_0)$. Find the ML estimate $\hat{c}$ of c in terms of the n samples x_i of x .

Solution. For n independent and identically distributed samples, the likelihood function is:

$$f(X, c) = \prod_{i=1}^n c e^{-c(x_i - x_0)} = c^n e^{-c \sum_{i=1}^n (x_i - x_0)}$$

The log-likelihood function is:

$$L(c) = n \ln c - c \sum_{i=1}^n (x_i - x_0)$$

Differentiating with respect to c and setting it to zero:

$$\frac{dL(c)}{dc} = \frac{n}{c} - \sum_{i=1}^n (x_i - x_0) = 0$$

Solving for c yields the ML estimate:

$$\hat{c} = \frac{n}{\sum_{i=1}^n (x_i - x_0)} = \frac{1}{\bar{x} - x_0}$$

where $\bar{x}$ is the sample mean. □

Problem 3

Problem Statement: Show that if $L(x, \theta) = \log f(x, \theta)$ is the likelihood function of a random variable x , then:

$$E \left\{ \left| \frac{\partial L(x, \theta)}{\partial \theta} \right|^2 \right\} = -E \left\{ \frac{\partial^2 L(x, \theta)}{\partial \theta^2} \right\}$$

Solution. Starting from the normalization condition $\int f(x, \theta) dx = 1$, we differentiate with respect to θ :

$$\int \frac{\partial f(x, \theta)}{\partial \theta} dx = 0$$

Using the identity $\frac{\partial L}{\partial \theta} = \frac{1}{f} \frac{\partial f}{\partial \theta}$, we have $\frac{\partial f}{\partial \theta} = f \frac{\partial L}{\partial \theta}$. Thus:

$$\int f \frac{\partial L}{\partial \theta} dx = E \left\{ \frac{\partial L}{\partial \theta} \right\} = 0$$

Differentiating the integral $\int \frac{\partial f}{\partial \theta} dx = 0$ a second time:

$$\int \frac{\partial^2 f(x, \theta)}{\partial \theta^2} dx = 0$$

The second derivative of L is given by:

$$\frac{\partial^2 L}{\partial \theta^2} = \frac{\partial}{\partial \theta} \left(\frac{1}{f} \frac{\partial f}{\partial \theta} \right) = \frac{1}{f} \frac{\partial^2 f}{\partial \theta^2} - \frac{1}{f^2} \left(\frac{\partial f}{\partial \theta} \right)^2 = \frac{1}{f} \frac{\partial^2 f}{\partial \theta^2} - \left(\frac{\partial L}{\partial \theta} \right)^2$$

Taking the expectation:

$$E \left\{ \frac{\partial^2 L}{\partial \theta^2} \right\} = \int f \left[\frac{1}{f} \frac{\partial^2 f}{\partial \theta^2} - \left(\frac{\partial L}{\partial \theta} \right)^2 \right] dx = \int \frac{\partial^2 f}{\partial \theta^2} dx - E \left\{ \left| \frac{\partial L}{\partial \theta} \right|^2 \right\}$$

Since $\int \frac{\partial^2 f}{\partial \theta^2} dx = 0$, the relationship simplifies to:

$$E \left\{ \frac{\partial^2 L}{\partial \theta^2} \right\} = -E \left\{ \left| \frac{\partial L}{\partial \theta} \right|^2 \right\}$$

Multiplying by -1 completes the proof. $\square$

Problem 4

Problem Statement: A complex discrete-time WSS process $z(\cdot)$ has mean zero and spectral density S_z . A new process $w(\cdot)$ is defined by $w(k) = z(k)e^{i\theta}$ where $\theta \sim U[0, 2\pi]$ is independent of $z(\cdot)$. Show that $S_w(\omega) = S_z(\omega)$ for all ω .

Solution. To show that $S_w(\omega) = S_z(\omega)$, we first verify that the process $w(k)$ is Wide-Sense Stationary (WSS) and then calculate its spectral density.

1. Mean of $w(k)$

The mean of $w(k)$ is given by:

$$\mu_w(k) = E[w(k)] = E[z(k)e^{i\theta}]$$

Since $z(\cdot)$ and θ are independent, the expectation of their product is the product of their expectations:

$$\mu_w(k) = E[z(k)]E[e^{i\theta}]$$

Given that $z(\cdot)$ has mean zero ($\mu_z = 0$), we have:

$$\mu_w(k) = 0 \cdot E[e^{i\theta}] = 0$$

Thus, μ_w is constant and equal to zero.

2. Autocovariance of $w(k)$

The autocovariance of $w(k)$ for lag m is defined as:

$$C_w(k, k+m) = E[w(k)w^*(k+m)] - E[w(k)]E[w^*(k+m)]$$

Since the mean is zero, this simplifies to the autocorrelation:

$$C_w(k, k+m) = E[(z(k)e^{i\theta})(z(k+m)e^{i\theta})^*]$$

$$C_w(k, k+m) = E[z(k)z^*(k+m)e^{i\theta}e^{-i\theta}]$$

$$C_w(k, k+m) = E[z(k)z^*(k+m) \cdot 1]$$

By independence:

$$C_w(k, k+m) = E[z(k)z^*(k+m)]$$

Because $z(\cdot)$ is WSS, its autocovariance depends only on the lag $m = (k+m) - k$

$$C_w(m) = C_z(m)$$

3. Spectral Density

The spectral density $S_w(\omega)$ is defined as the Discrete-Time Fourier Transform (DTFT) of the autocovariance function $C_w(m)$

$$S_w(\omega) = \sum_{m=-\infty}^{\infty} C_w(m) e^{-i\omega m}$$

Substituting $C_w(m) = C_z(m)$ into the definition:

$$S_w(\omega) = \sum_{m=-\infty}^{\infty} C_z(m) e^{-i\omega m} = S_z(\omega)$$

Therefore, $S_w(\omega) = S_z(\omega)$ for all ω . □

Problem 5

Problem Statement: A discrete-time white process $x(\cdot)$ with variance σ^2 is passed through a stable discrete-time LTI system with impulse response $h(\cdot)$. Find the variance of the output process $y(\cdot)$ in terms of σ^2 and h .

Solution. Let $x(k)$ be a white process. Its autocovariance is given by $C_x(k) = \sigma^2 \delta(k)$. The output of the LTI system is $y(k) = \sum_n h(n)x(k-n)$.

The variance of the output process is its autocovariance at lag zero, $C_y(0)$. Using the relationship for WSS processes through LTI systems:

$$C_y(k) = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} h(n)h(m)C_x(k-n+m)$$

Setting $k = 0$:

$$Var(y) = C_y(0) = \sum_n \sum_m h(n)h(m)\sigma^2 \delta(m-n)$$

Due to the sifting property of the Kronecker delta, the sum over m is non-zero only when $m = n$:

$$Var(y) = \sigma^2 \sum_{n=-\infty}^{\infty} h(n)h(n) = \sigma^2 \sum_{n=-\infty}^{\infty} |h(n)|^2$$

Thus, the output variance is the input variance scaled by the energy of the system's impulse response. □

Problem 6

Problem Statement: A zero-mean WSS discrete-time process $y(\cdot)$ has spectral density:

$$S_y(\omega) = \frac{5 + 3\cos(\omega)}{17 + 8\cos(\omega)}$$

- (a) Find the frequency response $H(\omega)$ of a discrete-time LTI system that will produce a process with this spectral density from a unit-variance white input signal $x(\cdot)$.
- (b) Calculate the autocovariance function $c_y(\kappa)$ for $\kappa \geq 0$.
- (c) Find the impulse response of a whitening filter for $y(\cdot)$.

Solution. Part (a)

We use spectral factorization. Given $S_x(\omega) = 1$, we have $S_y(\omega) = |H(\omega)|^2$. Convert to the z -domain by letting $e^{j\omega} = z$ and $\cos(\omega) = \frac{1}{2}(z + z^{-1})$:

$$S_y(z) = \frac{5 + 1.5(z + z^{-1})}{17 + 4(z + z^{-1})} = \frac{1.5z + 5 + 1.5z^{-1}}{4z + 17 + 4z^{-1}} = \frac{1.5(z + 3)(z + 1/3)}{4(z + 4)(z + 1/4)}$$

To ensure stability and causality, we choose the poles and zeros inside the unit circle ($|z| < 1$):

$$S_y(z) = \underbrace{\frac{\sqrt{4.5}(1 + \frac{1}{3}z^{-1})}{4(1 + \frac{1}{4}z^{-1})}}_{H(z)} \cdot \underbrace{\frac{\sqrt{4.5}(1 + \frac{1}{3}z)}{4(1 + \frac{1}{4}z)}}_{H(z^{-1})}$$

$$\text{Thus, } H(\omega) = \frac{3}{4\sqrt{2}} \frac{1 + \frac{1}{3}e^{-j\omega}}{1 + \frac{1}{4}e^{-j\omega}}.$$

Part (b)

The process is an ARMA(1,1) process defined by $y(k) + \frac{1}{4}y(k-1) = b_0x(k) + b_1x(k-1)$ with $b_0 = \frac{3}{4\sqrt{2}}$ and $b_1 = \frac{1}{4\sqrt{2}}$. The variance $c_y(0)$ is:

$$c_y(0) = \frac{b_0^2 + b_1^2 - 2ab_0b_1}{1 - a^2} = \frac{9/32 + 1/32 - 2(1/4)(3/32)}{1 - 1/16} = \frac{17/60}{1 - 1/16}$$

For $\kappa \geq 1$:

$$c_y(1) = b_0b_1 - ac_y(0) = \frac{3}{32} - \frac{1}{4} \left(\frac{17}{60} \right) = \frac{11}{480}$$

For $\kappa > 1$, $c_y(\kappa) = (-\frac{1}{4})c_y(\kappa-1)$. Thus:

$$c_y(\kappa) = \frac{11}{480} \left(-\frac{1}{4} \right)^{\kappa-1}, \quad \kappa \geq 1$$

Part (c)

The whitening filter $H_w(z)$ is the inverse of $H(z)$:

$$H_w(z) = \frac{1}{H(z)} = \frac{4\sqrt{2}}{3} \frac{1 + \frac{1}{4}z^{-1}}{1 + \frac{1}{3}z^{-1}}$$

Performing partial fraction expansion:

$$H_w(z) = \frac{4\sqrt{2}}{3} \left[1 - \frac{1/12 \cdot z^{-1}}{1 + \frac{1}{3}z^{-1}} \right]$$

The impulse response is:

$$h_w(k) = \frac{4\sqrt{2}}{3} \left(\delta(k) - \frac{1}{12} \left(-\frac{1}{3} \right)^{k-1} u(k-1) \right)$$

□

Problem 7

Problem Statement: With $\theta \sim U[0, 2\pi)$, show that the continuous-time process $x(t) = \cos(t + \theta)$ is Wide-Sense Stationary (WSS). Find the spectral density S_x in terms of the Dirac delta symbol.

Solution. To show that $x(t)$ is WSS, we must verify that its mean is constant and its autocovariance depends only on the time difference τ .

1. Mean Analysis

The mean of $x(t)$ is calculated by taking the expectation over the random variable θ :

$$E[x(t)] = E[\cos(t + \theta)] = \int_0^{2\pi} \cos(t + \theta) \frac{1}{2\pi} d\theta$$

Since we are integrating a cosine function over exactly one full period (2π), the result is zero:

$$E[x(t)] = \frac{1}{2\pi} [\sin(t + \theta)]_0^{2\pi} = \frac{1}{2\pi} (\sin(t + 2\pi) - \sin(t)) = 0$$

The mean is a constant (0), satisfying the first condition of WSS.

2. Autocovariance Analysis

The autocovariance (which is equal to the autocorrelation here since the mean is zero) is:

$$C_x(t, t + \tau) = E[x(t)x(t + \tau)] = E[\cos(t + \theta) \cos(t + \tau + \theta)]$$

Using the trigonometric identity $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$:

$$C_x(t, t + \tau) = \frac{1}{2} E[\cos((t + \tau + \theta) - (t + \theta)) + \cos((t + \tau + \theta) + (t + \theta))]$$

$$C_x(t, t + \tau) = \frac{1}{2} E[\cos(\tau) + \cos(2t + \tau + 2\theta)]$$

Applying the linearity of expectation:

$$C_x(t, t + \tau) = \frac{1}{2} \cos(\tau) + \frac{1}{2} E[\cos(2t + \tau + 2\theta)]$$

The second term involves an integral of a cosine over two full periods (due to the 2θ), which evaluates to zero. Thus:

$$C_x(\tau) = \frac{1}{2} \cos(\tau)$$

Since the mean is constant and the autocovariance depends only on τ , $x(t)$ is ****WSS****.

3. Spectral Density

The spectral density $S_x(\omega)$ is the Fourier Transform of the autocovariance $C_x(\tau)$:

$$S_x(\omega) = \int_{-\infty}^{\infty} C_x(\tau) e^{-j\omega\tau} d\tau = \int_{-\infty}^{\infty} \frac{1}{2} \cos(\tau) e^{-j\omega\tau} d\tau$$

Using the Euler identity $\cos(\tau) = \frac{1}{2}(e^{j\tau} + e^{-j\tau})$:

$$S_x(\omega) = \frac{1}{4} \int_{-\infty}^{\infty} (e^{j\tau} + e^{-j\tau}) e^{-j\omega\tau} d\tau$$

$$S_x(\omega) = \frac{1}{4} \left[\int_{-\infty}^{\infty} e^{-j(\omega-1)\tau} d\tau + \int_{-\infty}^{\infty} e^{-j(\omega+1)\tau} d\tau \right]$$

Using the property of the Dirac delta $\int_{-\infty}^{\infty} e^{-j\Omega t} dt = 2\pi\delta(\Omega)$:

$$S_x(\omega) = \frac{1}{4} [2\pi\delta(\omega-1) + 2\pi\delta(\omega+1)]$$

$$S_x(\omega) = \frac{\pi}{2} [\delta(\omega-1) + \delta(\omega+1)]$$

□

Problem 8

Problem Statement: Independent random variables ξ and η both have mean zero and variance σ^2 . Find the autocovariance function of the continuous-time process

$$x(t) = \xi \cos(\omega t) + \eta \sin(\omega t)$$

for fixed ω .

Solution. To find the autocovariance function $C_x(t_1, t_2)$, we first determine the mean of the process and then compute the expectation of the product of the process at two different times.

1. Mean of the Process

The mean $E[x(t)]$ is given by:

$$E[x(t)] = E[\xi \cos(\omega t) + \eta \sin(\omega t)]$$

Using the linearity of expectation:

$$E[x(t)] = E[\xi] \cos(\omega t) + E[\eta] \sin(\omega t)$$

Since ξ and η both have mean zero ($E[\xi] = E[\eta] = 0$):

$$E[x(t)] = 0 \cdot \cos(\omega t) + 0 \cdot \sin(\omega t) = 0$$

The mean is constant at zero.

2. Autocovariance Calculation

Since the mean is zero, the autocovariance $C_x(t_1, t_2)$ is equal to the autocorrelation $R_x(t_1, t_2)$:

$$C_x(t_1, t_2) = E[x(t_1)x(t_2)]$$

Substituting the definition of $x(t)$:

$$C_x(t_1, t_2) = E[(\xi \cos(\omega t_1) + \eta \sin(\omega t_1))(\xi \cos(\omega t_2) + \eta \sin(\omega t_2))]$$

Expanding the product:

$$\begin{aligned} C_x(t_1, t_2) = & E[\xi^2 \cos(\omega t_1) \cos(\omega t_2) + \xi \eta \cos(\omega t_1) \sin(\omega t_2) \\ & + \eta \xi \sin(\omega t_1) \cos(\omega t_2) + \eta^2 \sin(\omega t_1) \sin(\omega t_2)] \end{aligned}$$

Applying the linearity of expectation and using the fact that ξ and η are independent ($E[\xi\eta] = E[\xi]E[\eta] = 0$):

$$C_x(t_1, t_2) = E[\xi^2] \cos(\omega t_1) \cos(\omega t_2) + E[\eta^2] \sin(\omega t_1) \sin(\omega t_2)$$

Given $E[\xi^2] = E[\eta^2] = \sigma^2$:

$$C_x(t_1, t_2) = \sigma^2 (\cos(\omega t_1) \cos(\omega t_2) + \sin(\omega t_1) \sin(\omega t_2))$$

Using the trigonometric identity $\cos(A - B) = \cos A \cos B + \sin A \sin B$:

$$C_x(t_1, t_2) = \sigma^2 \cos(\omega(t_1 - t_2))$$

3. Conclusion

Letting $\tau = t_1 - t_2$, we see that the autocovariance depends only on the time difference:

$$C_x(\tau) = \sigma^2 \cos(\omega\tau)$$

This result shows that the process is Wide-Sense Stationary (WSS), with its "power" concentrated at the frequency ω . $\square$

Problem 9

Problem Statement: A continuous-time process $x(\cdot)$ is WSS with mean zero and spectral density

$$S_x(\omega) = \begin{cases} a^2 - \omega^2, & |\omega| \leq a \\ 0, & |\omega| > a \end{cases}$$

Calculate $Var(x(t))$.

Solution. For a Wide-Sense Stationary (WSS) process, the variance is equal to the autocovariance function at lag zero, $C_x(0)$. The relationship between $C_x(\tau)$ and the power spectral density $S_x(\omega)$ is defined by the Inverse Fourier Transform:

$$C_x(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_x(\omega) e^{j\omega\tau} d\omega$$

To find the variance, we set the time lag $\tau = 0$:

$$\text{Var}(x(t)) = C_x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_x(\omega) d\omega$$

Substituting the given spectral density into the integral:

$$\text{Var}(x(t)) = \frac{1}{2\pi} \int_{-a}^a (a^2 - \omega^2) d\omega$$

Since the integrand $a^2 - \omega^2$ is an even function, we can compute the integral over the interval $[0, a]$ and multiply by 2:

$$\text{Var}(x(t)) = \frac{2}{2\pi} \int_0^a (a^2 - \omega^2) d\omega$$

$$\text{Var}(x(t)) = \frac{1}{\pi} \left[a^2\omega - \frac{\omega^3}{3} \right]_0^a$$

Evaluating at the boundaries:

$$\text{Var}(x(t)) = \frac{1}{\pi} \left((a^2 \cdot a - \frac{a^3}{3}) - 0 \right)$$

$$\text{Var}(x(t)) = \frac{1}{\pi} \left(a^3 - \frac{a^3}{3} \right) = \frac{1}{\pi} \left(\frac{2a^3}{3} \right)$$

The final result for the variance is:

$$\text{Var}(x(t)) = \frac{2a^3}{3\pi}$$

□

Problem 10

Problem Statement: The spectral density of a continuous-time WSS process $x(\cdot)$ is

$$S_x(\omega) = \begin{cases} 1 - \frac{|\omega|}{W}, & |\omega| \leq W \\ 0, & |\omega| > W \end{cases}$$

What is the autocovariance function of this process?

Solution. The autocovariance function $C_x(\tau)$ is the inverse Fourier transform of the power spectral density $S_x(\omega)$:

$$C_x(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_x(\omega) e^{j\omega\tau} d\omega$$

Substituting the given triangular spectral density:

$$C_x(\tau) = \frac{1}{2\pi} \int_{-W}^W \left(1 - \frac{|\omega|}{W}\right) e^{j\omega\tau} d\omega$$

Since $S_x(\omega)$ is an even function of ω , the imaginary part of the integral (sine term) vanishes:

$$C_x(\tau) = \frac{1}{\pi} \int_0^W \left(1 - \frac{\omega}{W}\right) \cos(\omega\tau) d\omega$$

Using integration by parts, let $u = 1 - \frac{\omega}{W}$ and $dv = \cos(\omega\tau) d\omega$:

$$C_x(\tau) = \frac{1}{\pi} \left[\left(1 - \frac{\omega}{W}\right) \frac{\sin(\omega\tau)}{\tau} \right]_0^W + \frac{1}{\pi W \tau} \int_0^W \sin(\omega\tau) d\omega$$

Evaluating the terms:

$$C_x(\tau) = 0 + \frac{1}{\pi W \tau} \left[-\frac{\cos(\omega\tau)}{\tau} \right]_0^W = \frac{1 - \cos(W\tau)}{\pi W \tau^2}$$

Using the identity $1 - \cos(2\theta) = 2 \sin^2(\theta)$:

$$C_x(\tau) = \frac{2 \sin^2(W\tau/2)}{\pi W \tau^2} = \frac{W}{2\pi} \left(\frac{\sin(W\tau/2)}{W\tau/2} \right)^2$$

□

Problem 11

Problem Statement: A continuous-time stochastic process $x(\cdot)$ is said to have independent increments if, for all $t_1 < t_2 < t_3 < t_4$, $x(t_4) - x(t_3)$ is independent of $x(t_2) - x(t_1)$. For such a process, determine $c_x(t_1, t_2)$ for $0 \leq t_1 < t_2$ in terms of $\text{var}(x(t_1))$ assuming $x(0) = 0$.

Solution. The autocovariance function $c_x(t_1, t_2)$ is defined as the covariance between the random variables at two different times:

$$c_x(t_1, t_2) = \text{cov}(x(t_1), x(t_2))$$

For the interval $0 \leq t_1 < t_2$, we can express the value of the process at the later time t_2 as the sum of its value at t_1 and the change (increment) that occurred between t_1 and t_2 :

$$x(t_2) = x(t_1) + [x(t_2) - x(t_1)]$$

Substituting this expression into the covariance formula:

$$c_x(t_1, t_2) = \text{cov}(x(t_1), x(t_1) + [x(t_2) - x(t_1)])$$

By the linearity property of covariance, specifically $\text{cov}(A, B+C) = \text{cov}(A, B) + \text{cov}(A, C)$, we can split the terms:

$$c_x(t_1, t_2) = \text{cov}(x(t_1), x(t_1)) + \text{cov}(x(t_1), x(t_2) - x(t_1))$$

The first term is the covariance of a variable with itself, which is its variance:

$$\text{cov}(x(t_1), x(t_1)) = \text{var}(x(t_1))$$

To evaluate the second term, we use the provided assumption $x(0) = 0$. This allows us to write $x(t_1)$ as the increment $x(t_1) - x(0)$. Since $0 \leq 0 < t_1 < t_2$, the property of independent increments states that the increment $[x(t_1) - x(0)]$ is independent of the increment $[x(t_2) - x(t_1)]$.

Since independent random variables have zero covariance:

$$\text{cov}(x(t_1), x(t_2) - x(t_1)) = 0$$

Summing the results, we find:

$$c_x(t_1, t_2) = \text{var}(x(t_1)) + 0 = \text{var}(x(t_1))$$

Thus, for $0 \leq t_1 < t_2$, the autocovariance is simply the variance at the earlier time point. $\square$

Problem 12

Problem Statement: Using 2048 samples from `wss.dat` and a sampling rate of 5 MHz, estimate the spectral density S_x using two specific methods: (a) Averaging magnitude-squared 128-point FFTs of 16 non-overlapping blocks. (b) Taking the magnitude-squared FFT of a 128-point symmetric autocovariance sequence.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 data = np.loadtxt("wss.dat")
5 fs = 5.0
6 N_fft = 128
7
8 freq = -2.5 + (5 / 128) * np.arange(1, 129)
9
10 pg = np.zeros(N_fft)
11 for j in range(16):
12     block = data[j * 128 : (j + 1) * 128]
13     ffts = np.fft.fft(block)
14     pg += np.abs(np.fft.fftshift(ffts)) ** 2
15

```

```

16
17 def estimate_autocovariance(x, max_lag):
18     n = len(x)
19     x_centered = x - np.mean(x)
20     return np.array(
21         [np.mean(x_centered[: n - k] * x_centered[k:]) for k in
22          range(max_lag + 1)]
23     )
24
25 ac = estimate_autocovariance(data, 64)
26 acn = ac[::-1]
27 acs = np.concatenate([acn, ac[1:64]])
28
29 psd_b = np.abs(np.fft.fftshift(np.fft.fft(acs))) ** 2
30
31 plt.figure(figsize=(12, 5))
32
33 plt.subplot(1, 2, 1)
34 plt.plot(freq, pg)
35 plt.title("Method (a): Averaged Periodograms")
36 plt.xlabel("Frequency (MHz)")
37 plt.ylabel("Estimated spectral density")
38 plt.grid(True)
39
40 plt.subplot(1, 2, 2)
41 plt.plot(freq, psd_b)
42 plt.title("Method (b): FFT of Autocovariance")
43 plt.xlabel("Frequency (MHz)")
44 plt.ylabel("Estimated spectral density")
45 plt.grid(True)
46
47 plt.tight_layout()
48 plt.show()

```

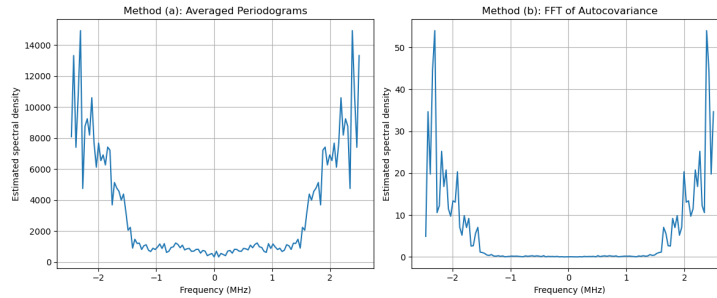


Figure 1: Estimated spectral density S_x using the two methods.